

An AI-Augmented Framework for Validating Phenomenological Models of Particle Spectra

Candidate Name

July 9, 2026

Abstract

This thesis presents an autonomous, multi-agent AI framework designed to rigorously validate phenomenological physics models and literature claims. The framework is applied to evaluate a proposed 3-component model for transverse momentum (p_T) spectra of bosons produced in high-energy collisions. We definitively prove the intrinsic over-parameterization of the 2-component Jüttner model for high-multiplicity p_T spectra, contrasting its failure to converge against robust Blast-Wave (BGBW) and Tsallis baselines. Our framework seamlessly integrates Retrieval-Augmented Generation (RAG) and Model Context Protocol (MCP) toolchains, bridging the gap between symbolic validation, numerical fitting, and literature consensus.

Contents

1	Introduction	3
2	Theoretical Background	4
2.1	Transverse Momentum Spectra in High-Energy Collisions	4
2.2	The Blast-Wave Model (BGBW)	4
2.3	The Tsallis Distribution	4
2.4	The Jüttner/Boltzmann Approximation	5
3	Literature Review	6
3.1	Thermal and Hydrodynamical Foundations	6
3.2	The Tsallis Distribution and Non-Extensive Thermodynamics	6
3.3	Recent Advancements and Multiplicity Dependence	7
4	AI Methodology	8
4.1	System Architecture and Orchestration	8
4.2	Retrieval-Augmented Generation for Scientific Validation	8
4.3	Automated Symbolic and Numerical Pipeline	8
5	Results and Validation	10
5.1	Baseline Validations: BGBW and Tsallis Models	10
5.2	Evaluation of the Single-Component Jüttner Model	10
5.3	Intrinsic Over-parameterization of the 2-Component Jüttner Model	11
6	Conclusion	12

Chapter 1

Introduction

The study of transverse momentum (p_T) spectra of particles produced in high-energy proton-proton and heavy-ion collisions provides critical insights into the thermodynamic properties and collective dynamics of the strongly interacting medium. A fundamental challenge in modeling these spectra lies in the tension between two distinct physical regimes: the soft "bulk" particle production at low p_T (dominated by thermal emission and collective hydrodynamic expansion) and the hard scattering tail at high p_T (dominated by perturbative quantum chromodynamics and jet fragmentation).

Historically, phenomenological models such as the Boltzmann-Gibbs Blast-Wave (BGBW) framework and the thermodynamically consistent Tsallis distribution have been employed to reconcile these regimes, extracting kinetic freeze-out temperatures and transverse flow velocities. However, attempts to unify these regimes have led to the proposition of increasingly complex multi-component models.

This thesis investigates one such proposal: a three-component model aiming to describe the full p_T spectrum using multiple moving Jüttner (relativistic Maxwell-Boltzmann) distributions. Evaluating the validity of such highly parameterized models requires moving beyond simple visual fits to rigorous statistical and physical verification.

To this end, we introduce a novel methodological contribution: an autonomous, multi-agent artificial intelligence framework designed to systematically validate phenomenological physics models against both experimental data and literature consensus. By applying this AI-augmented framework to the proposed Jüttner model, we execute an end-to-end validation pipeline to definitively assess its statistical rigor, physical justification, and structural stability against established literature baselines.

Chapter 2

Theoretical Background

2.1 Transverse Momentum Spectra in High-Energy Collisions

The invariant yield of produced particles is typically expressed as a function of transverse momentum p_T and rapidity y . For massless particles or at high p_T , the pseudorapidity η is often used as a convenient experimental approximation. The invariant cross section is given by:

$$E \frac{d^3 N}{dp^3} = \frac{d^2 N}{2\pi p_T dp_T dy} \quad (2.1)$$

2.2 The Blast-Wave Model (BGBW)

The Boltzmann-Gibbs Blast-Wave (BGBW) model assumes a locally thermalized source expanding with a collective transverse velocity profile. The invariant yield is obtained by integrating the thermal distribution over the expanding freeze-out hyper-surface:

$$\frac{d^2 N}{dp_T dy} \propto p_T m_T \int_0^R r dr I_0 \left(\frac{p_T \sinh \rho}{T_{\text{kin}}} \right) K_1 \left(\frac{m_T \cosh \rho}{T_{\text{kin}}} \right) \quad (2.2)$$

where $m_T = \sqrt{p_T^2 + m^2}$ is the transverse mass, T_{kin} is the kinetic freeze-out temperature, $\rho = \tanh^{-1} \beta_T$ is the transverse rapidity, and $\beta_T(r) = \beta_s(r/R)^n$ is the transverse velocity profile. The parameters I_0 and K_1 are modified Bessel functions.

2.3 The Tsallis Distribution

The non-extensive Tsallis distribution accounts for temperature fluctuations and hard scattering power-law tails at high p_T . The thermodynamically consistent form, ensuring standard thermodynamic relations for pressure and energy density, is formulated as

[Cleymans and Worku, 2012]:

$$\frac{d^2 N}{dp_T dy} = gV \frac{p_T m_T}{(2\pi)^2} \left[1 + (q-1) \frac{m_T}{T} \right]^{-\frac{q}{q-1}} \quad (2.3)$$

where q is the non-extensivity parameter. As $q \rightarrow 1$, this form recovers the standard Boltzmann-Gibbs exponential distribution.

2.4 The Jüttner/Boltzmann Approximation

For large transverse masses or temperatures where quantum degeneracy is negligible, the full Bose-Einstein or Fermi-Dirac statistics approach the classical relativistic Maxwell-Boltzmann (Jüttner) distribution. The phase-space density in a fluid cell moving with four-velocity U^μ is given by:

$$f(p) \propto \exp\left(-\frac{p^\mu U_\mu}{T}\right) \quad (2.4)$$

For a static system ($U = (1, 0, 0, 0)$), this reduces to $f(p) \propto \exp(-E/T)$. In models proposing multiple moving thermal sources, the argument becomes $p^\mu U_\mu = m_T \cosh(y - Y_{source}) - p_T \cos(\phi) U_T$. Over-parameterization of U^μ often leads to severe structural instability in numerical fits.

Chapter 3

Literature Review

The study of transverse momentum (p_T) spectra in high-energy proton-proton and heavy-ion collisions has a rich history, with phenomenological models continually evolving to capture both the soft (thermal) and hard (perturbative QCD) regimes of particle production. This chapter reviews the foundational models and recent advancements that form the baseline for our validation framework.

3.1 Thermal and Hydrodynamical Foundations

Early approaches to modeling particle spectra relied heavily on the Boltzmann-Gibbs statistics, assuming local thermal equilibrium of the produced fireball. However, the observation of power-law tails at high p_T necessitated refinements. The Blast-Wave model [Schnedermann et al., 1993] introduced collective transverse flow, successfully describing the soft bulk of the spectra ($p_T \lesssim 2.5$ GeV/c) by assuming a locally thermalized source expanding with a collective transverse velocity profile.

The standard Boltzmann-Gibbs Blast-Wave (BGBW) model evaluates the invariant yield as an integral over the freeze-out surface. While highly successful at low p_T , the BGBW model systematically underpredicts the yield at higher transverse momenta, where hard parton scattering dominates.

3.2 The Tsallis Distribution and Non-Extensive Thermodynamics

To simultaneously describe both the low- p_T thermal bulk and the high- p_T power-law tail, the non-extensive Tsallis statistics were introduced to high-energy physics. A seminal advancement was the formulation of a thermodynamically consistent Tsallis distribution by Cleymans and Worku [2012], which strictly satisfies thermodynamic relations for particle number, energy density, and pressure. The Tsallis distribution introduces a

non-extensivity parameter q ; as $q \rightarrow 1$, the distribution recovers the standard Boltzmann-Gibbs form.

3.3 Recent Advancements and Multiplicity Dependence

Primary measurements of charged-particle p_T spectra at the LHC have been extensively provided by the ALICE Collaboration [ALICE Collaboration, 2018]. These foundational datasets spanning multiple collision systems (pp , p -Pb, and Pb-Pb) and collision energies (up to $\sqrt{s} = 13$ TeV) serve as the empirical bedrock for phenomenological modeling.

Recent studies have focused heavily on the multiplicity dependence of freeze-out parameters in these proton-proton collisions. Khuntia et al. [2019] analyzed p_T spectra at $\sqrt{s} = 7$ TeV using both the BGBW and the thermodynamically consistent Tsallis distribution. They observed a mass ordering in the kinetic freeze-out temperature T_{kin} and the non-extensivity parameter, supporting a differential freeze-out scenario. Crucially, their BGBW analysis showed that the average radial flow velocity $\langle\beta\rangle$ remains largely independent of the event multiplicity, providing a key benchmark for multi-component models.

Rath et al. [2020] extended this work to 5.02 and 13 TeV collision energies. Their findings corroborated that $\langle\beta\rangle$ is almost independent of collision energy and multiplicity. However, they found that T_{kin} significantly depends on the multiplicity classes. They concluded that while the Tsallis model better describes the full p_T range, the BGBW model remains highly effective for the bulk part ($p_T \lesssim 2.5$ GeV/c).

More recently, Bíró et al. [2025] proposed a two-component soft/hard baseline model for ALICE pp data from 2.76 to 13 TeV. Their analysis revealed that a simple Boltzmann fit effectively describes the soft component across all energies, while the residual hard spectra exhibit no evolution in shape or peak position with event multiplicity. Furthermore, the mean p_T for both the soft and hard components remains nearly constant across multiplicity classes, challenging more complex hydrodynamical interpretations that necessitate highly parameterized models.

These literature benchmarks underscore the necessity of rigorous statistical validation when proposing novel multi-component models, such as the 3-component formulation (incorporating a moving Jüttner distribution) evaluated in this thesis.

Chapter 4

AI Methodology

The core methodological contribution of this thesis is the design and deployment of an autonomous, multi-agent AI framework capable of end-to-end scientific validation. This framework transforms a traditional static code repository into a dynamic "validation workspace."

4.1 System Architecture and Orchestration

The framework is orchestrated using the DeerFlow platform, which coordinates multi-agent reasoning. Agents are equipped with specific toolchains via the Model Context Protocol (MCP), allowing them to interface with both local execution environments and remote scientific databases. This enables a seamless transition from literature retrieval to symbolic mathematics and numerical fitting.

4.2 Retrieval-Augmented Generation for Scientific Validation

A central component of our methodology is an Onyx-backed Retrieval-Augmented Generation (RAG) system. This system interfaces directly with academic APIs, including Semantic Scholar, arXiv, and Scite. By automating citation network analysis, the framework can evaluate the consensus around specific physical assumptions (e.g., the validity of the $\eta \rightarrow \gamma$ Jacobian transformation at high p_T). This ensures that the baseline formulas used in our validation are historically and statistically grounded.

4.3 Automated Symbolic and Numerical Pipeline

The computational validation is split into two layers:

1. **Symbolic Validation:** The framework parses draft manuscript equations and uses SymPy to symbolically verify Lorentz covariance, integration limits, and thermodynamic consistency (e.g., recovering the Cooper-Frye static limit as flow velocity $U \rightarrow 0$).
2. **Numerical Fitting:** After symbolic verification, the equations are compiled into a numerical pipeline (`fitting_pipeline.py`) utilizing the `iminuit` optimizer. The pipeline is designed to enforce rigorous statistical checks, including the computation of the covariance matrix, parameter correlations, residuals, and χ^2/ndf .

This dual-layer approach prevents unphysical models (e.g., those suffering from intrinsic over-parameterization) from being accepted based on superficially reasonable visual fits.

Chapter 5

Results and Validation

This chapter presents the numerical outcomes of the validation pipeline when applied to p_T spectra measured in high-energy collisions. The pipeline systematically evaluates single-component and multi-component models across ten multiplicity classes.

5.1 Baseline Validations: BGBW and Tsallis Models

To establish a rigorous baseline, we first fitted the spectra using the single-component BGBW and thermodynamically consistent Tsallis distributions.

The BGBW model yielded physically interpretable freeze-out parameters that perfectly align with expected literature trends (e.g., Khuntia et al. [2019], Rath et al. [2020]). At low multiplicity (class 21-30), the kinetic freeze-out temperature was found to be $T_{\text{kin}} = 132$ MeV with an average transverse flow velocity $\langle\beta\rangle = 0.31$. At high multiplicity (class 126-150), T_{kin} decreased to 87 MeV while the flow velocity increased to $\langle\beta\rangle = 0.66$. The fit quality was consistently excellent, with χ^2/ndf ranging strictly between 1.12 and 2.06 across all multiplicity bins.

The Tsallis baseline also performed well, though with slightly higher variance in fit quality (χ^2/ndf between 0.33 and 14.29).

5.2 Evaluation of the Single-Component Jüttner Model

In contrast to the BGBW model, the single-component Jüttner (Maxwell-Boltzmann) model completely failed to describe the data. The χ^2/ndf values ranged from 50.1 at low multiplicity to an extreme 176.8 at high multiplicity. This confirms that a simple exponential distribution without collective flow or hard scattering tails cannot adequately capture the p_T spectra dynamics.

5.3 Intrinsic Over-parameterization of the 2-Component Jüttner Model

The central claim of the evaluated manuscript proposed a multi-component model relying on moving Jüttner distributions. Our validation pipeline tested a 2-component formulation of this model to evaluate its structural stability.

The `iminuit` optimizer failed to converge in 9 out of 10 multiplicity bins, returning invalid covariance matrices and Expected Distance to Minimum (EDM) values > 0.1 . In a properly constrained model, the negative log-likelihood (or χ^2) surface exhibits a well-defined parabolic minimum, allowing the Hessian matrix to be inverted to calculate parameter uncertainties. However, when a model is over-parameterized—as in the case of multiple unconstrained four-velocity vectors U^μ —the likelihood manifold becomes extremely flat in certain directions. This manifests as highly correlated parameter pairs (with off-diagonal elements in the correlation matrix approaching ± 1.0) and physically meaningless, diverging parameter uncertainties.

An exhaustive dense grid scan over the initial parameter space (T, U_1, U_2) confirmed that this convergence failure is not due to a poor initial guess (i.e., a local minimum trap), but rather an intrinsic topological feature of the over-parameterized model space. Even in isolated scenarios where the optimizer was forced to return a "success" state by artificially tightening parameter bounds, the resulting χ^2/ndf exceeded 172.5. This indicates that even the "best" mathematical fit provides a physically unviable description of the experimental data.

We therefore definitively conclude that the proposed 2-component Jüttner model is structurally unstable and unsuited for describing the multiplicity evolution of p_T spectra.

Chapter 6

Conclusion

This thesis establishes that an autonomous, multi-agent AI framework can successfully execute a rigorous, end-to-end validation of complex phenomenological physics models. By systematically evaluating p_T spectra across ten multiplicity classes, we investigated the long-standing physical tension between soft thermal emission and hard scattering tails.

Our numerical pipeline exposed the intrinsic limitations of the proposed 2-component Jüttner model. The model’s failure to converge—driven by extreme parameter correlation and flat χ^2 likelihood manifolds—demonstrates that naïvely introducing additional moving thermal sources results in severe over-parameterization rather than physical insight. Conversely, our framework independently validated the Boltzmann-Gibbs Blast-Wave (BGBW) baseline, extracting freeze-out parameters that align perfectly with established ALICE experimental literature.

Ultimately, this work proves that resolving the soft/hard tension in high-energy collisions requires robust, mathematically stable formalisms (such as the Tsallis distribution or explicit soft/hard decompositions) rather than the proliferation of unconstrained thermal components. The AI-augmented methodology developed herein establishes a new paradigm for automated peer review, ensuring that phenomenological proposals meet rigorous standards of reproducibility and statistical stability in high-energy physics.

Bibliography

- ALICE Collaboration. Transverse momentum spectra and nuclear modification factors of charged particles in pp , p -pb and pb-pb collisions at the lhc. *Journal of High Energy Physics*, 2018(11):13, 2018. doi: 10.1007/JHEP11(2018)013.
- T. S. Bíró, G. Paić, and L. Serkin. Robust alternative to hydrodynamical interpretations of proton-proton collisions, 2025.
- J. Cleymans and D. Worku. Relativistic thermodynamics: Transverse momentum distributions in high-energy physics. *The European Physical Journal A*, 48(11):160, 2012. doi: 10.1140/epja/i2012-12160-0.
- A. Khuntia, H. Sharma, S. K. Tiwari, J. Cleymans, and R. Sahoo. Radial flow and differential freeze-out in proton-proton collisions at $\sqrt{s} = 7$ tev at the lhc. *The European Physical Journal A*, 55(1):3, 2019. doi: 10.1140/epja/i2019-12669-6.
- R. Rath, A. Khuntia, R. Sahoo, and J. Cleymans. Event multiplicity, transverse momentum and energy dependence of charged particle production, and system thermodynamics in pp collisions at the large hadron collider. *Journal of Physics G: Nuclear and Particle Physics*, 47(5):055111, 2020. doi: 10.1088/1361-6471/ab783b.
- E. Schnedermann, J. Sollfrank, and U. Heinz. Thermal phenomenology of hadrons from 200a gev s+s collisions. *Physical Review C*, 48(5):2462, 1993. doi: 10.1103/PhysRevC.48.2462.